

MAT 426, Exam 2 Review: QR Factorization via Householder

14. (a) Find the QR Factorization of $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & -2 & 4 \end{bmatrix}$ using Householder matrices.

Let $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$. Then,

$$\mathbf{u}_1 = \mathbf{x}_1 + \|\mathbf{x}_1\|\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \sqrt{5} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + \sqrt{5} \\ 0 \\ 2 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} A^{(1)} &= H_1 A = A - \frac{2}{\mathbf{u}_1^T \mathbf{u}_1} \mathbf{u}_1 (A^T \mathbf{u}_1)^T \\ &= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & -2 & 4 \end{bmatrix} - \frac{2}{(1 + \sqrt{5})^2 + 0 + 4} \begin{bmatrix} 1 + \sqrt{5} \\ 0 \\ 2 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 2 \\ 2 & 3 & -2 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 + \sqrt{5} \\ 0 \\ 2 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & -2 & 4 \end{bmatrix} - \frac{1}{5 + \sqrt{5}} \begin{bmatrix} 1 + \sqrt{5} \\ 0 \\ 2 \end{bmatrix} [5 + \sqrt{5} \quad -2 + 2\sqrt{5} \quad 9 + \sqrt{5}] \\ &= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & -2 & 4 \end{bmatrix} - \frac{1}{5 + \sqrt{5}} \begin{bmatrix} 10 + 6\sqrt{5} & 8 & 14 + 10\sqrt{5} \\ 0 & 0 & 0 \\ 10 + 2\sqrt{5} & -4 + 4\sqrt{5} & 18 + 2\sqrt{5} \end{bmatrix} \\ &= \begin{bmatrix} -2.2361 & 0.8944 & -4.0249 \\ 0 & 3 & 2 \\ 0 & -2.6833 & 0.8944 \end{bmatrix} \end{aligned}$$

Next, let $\mathbf{x}_2 = \begin{bmatrix} 3 \\ -2.6833 \end{bmatrix}$. Then,

$$\mathbf{u}_2 = \mathbf{x}_2 + \|\mathbf{x}_2\|\mathbf{e}_2 = \begin{bmatrix} 3 \\ -2.6833 \end{bmatrix} + 4.0249 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7.0249 \\ -2.6833 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \bar{A}^{(2)} &= \bar{H}_2 \bar{A}^{(1)} = \bar{A}^{(1)} - \frac{2}{\mathbf{u}_2^T \mathbf{u}_2} \mathbf{u}_2 ((\bar{A}^{(1)})^T \mathbf{u}_2)^T \\ &= \begin{bmatrix} 3 & 2 \\ -2.6833 & 0.8944 \end{bmatrix} - 0.0354 \begin{bmatrix} 7.0249 \\ -2.6833 \end{bmatrix} \left(\begin{bmatrix} 3 & -2.6833 \\ 2 & 0.8944 \end{bmatrix} \begin{bmatrix} 7.0249 \\ -2.6833 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 3 & 2 \\ -2.6833 & 0.8944 \end{bmatrix} - 0.0354 \begin{bmatrix} 7.0249 \\ -2.6833 \end{bmatrix} [28.2748 \quad 11.6499] \\ &= \begin{bmatrix} 3 & 2 \\ -2.6833 & 0.8944 \end{bmatrix} - \begin{bmatrix} 7.0355 & 2.8988 \\ -2.6833 & -1.1066 \end{bmatrix} \\ &= \begin{bmatrix} -4.0355 & -0.8988 \\ 0 & 2.001 \end{bmatrix} \end{aligned}$$

Therefore,

$$R = H_2 H_1 A = \begin{bmatrix} -2.2361 & 0.8944 & -4.0249 \\ 0 & -4.0355 & -0.8988 \\ 0 & 0 & 2.001 \end{bmatrix}$$

To find Q we need to compute $Q = (H_2 H_1)^{-1} = (H_2 H_1)^T = H_1 H_2$. First, note that

$$\begin{aligned} H_1 &= I - \frac{2}{\mathbf{u}_1^T \mathbf{u}_1} \mathbf{u}_1 \mathbf{u}_1^T \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{10 + 2\sqrt{5}} \begin{bmatrix} 1 + \sqrt{5} \\ 0 \\ 2 \end{bmatrix} \begin{bmatrix} 1 + \sqrt{5} & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -0.4472 & 0 & -0.8944 \\ 0 & 1 & 0 \\ -0.8944 & 0 & 0.4472 \end{bmatrix} \end{aligned}$$

Next, note that

$$\begin{aligned} \hat{H}^2 &= I - \frac{2}{\mathbf{u}_2^T \mathbf{u}_2} \mathbf{u}_2 \mathbf{u}_2^T \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.0354 \begin{bmatrix} 7.0249 \\ -2.6833 \end{bmatrix} \begin{bmatrix} 7.0249 & -2.6833 \end{bmatrix} \\ &= \begin{bmatrix} -0.7470 & 0.6673 \\ 0.6673 & 0.7451 \end{bmatrix} \end{aligned}$$

Therefore,

$$H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.7470 & 0.6673 \\ 0 & 0.6673 & 0.7451 \end{bmatrix}$$

Finally,

$$\begin{aligned} Q &= H_1 H_2 \\ &= \begin{bmatrix} -0.4472 & 0 & -0.8944 \\ 0 & 1 & 0 \\ -0.8944 & 0 & 0.4472 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.7470 & 0.6673 \\ 0 & 0.6673 & 0.7451 \end{bmatrix} \\ &= \begin{bmatrix} -0.4472 & -0.5968 & -0.6664 \\ 0 & -0.7470 & 0.6673 \\ -0.8944 & 0.2984 & 0.3332 \end{bmatrix} \end{aligned}$$